$Aborted

In[27]:=

```
W[x_] := Cos[(2 * n - 1) * Pi * x / (2 * L)]
W[x]
Integrate[W[x], {x, 0, L}]
Integrate[W[x] * W[x], {x, 0, L}]
```

Out[28]=

$$\mathrm{Cos}\left[\frac{(-1+2\,n)\,\pi\,x}{2\,L}\right]$$

Out[29]=

$$\frac{2\,L\,\mathrm{Cos}[n\,\pi]}{\pi-2\,n\,\pi}$$

Out[30]=

$$\frac{1}{2}\left(L+\frac{L\,\mathrm{Sin}[2\,n\,\pi]}{\pi-2\,n\,\pi}\right)$$

COMPARISION

In[27]:=

```
(* ------------------------------------------------------------ *)
(* CLEAR WORKSPACE *)
(* ------------------------------------------------------------ *)
ClearAll[]



(* ------------------------------------------------------------ *)
(* GIVEN PARAMETERS *)
(* ------------------------------------------------------------ *)

l = 2;            (* Length (m) *)
d = 0.005;          (* Diameter (m) *)
ρ = 7850;          (* Density (kg/m^3) *)
T = 1000;          (* Tension (N) *)
Q0 = 60;           (* Force amplitude (N) *)
Ω = 220;           (* Excitation frequency (rad/s) *)

A = Pi d^2/4;        (* Cross-sectional area *)
c = Sqrt[T/(ρ A)];      (* Wave speed *)
k = Ω/c;           (* Frequency parameter *)

(* ------------------------------------------------------------ *)
```

```
(* (b) ANALYTICAL SOLUTION *)
(* ---------------------------------------------------- *)

Xanalytical[x_] :=
 (Q0 / (ρ A Ω ^ 2)) * (Cos[k x] / Cos[k l] - 1)

wA[x_, t_] := Xanalytical[x] × Cos[Ω t]

(* ---------------------------------------------------- *)
(* (c) EIGENFUNCTION EXPANSION (10 TERMS) *)
(* ---------------------------------------------------- *)

ω[k_] := ((2 k - 1) Pi c) / (2 l)

W[k_, x_] := Cos[((2 k - 1) Pi x) / (2 l)]

α[k_] :=
 (2 * l * Q0) /
  ((2 * k - 1) * Pi * ρ * A * (ω[k] ^ 2 - Ω ^ 2))

XE[x_] := Sum[α[k] × W[k, x], {k, 1, 10}]

wE[x_, t_] := XE[x] × Cos[Ω t]

(* ---------------------------------------------------- *)
(* (d) GREEN'S FUNCTION METHOD *)
(* ---------------------------------------------------- *)

G[x_, xb_] :=
 Piecewise[{
   {Cos[k x] × Sin[k (l - xb)] / (k Cos[k l]), x < xb},
   {Cos[k xb] × Sin[k (l - x)] / (k Cos[k l]), x > xb}
  }]

XG[x_] := Integrate[G[x, xb] * (Q0 / (ρ A)), {xb, 0, l}]

wG[x_, t_] := XG[x] × Cos[Ω t]

(* ---------------------------------------------------- *)
(* (e) COMPARISON PLOTS *)
(* ---------------------------------------------------- *)
```

```
t1 = 0;
t2 = Pi/(4 Ω);
t3 = Pi/(2 Ω);

Plot[
 Evaluate[{wA[x, t1], wE[x, t1], wG[x, t1]}],
 {x, 0, l},
 PlotLegends → {"Analytical", "Eigen (10 terms)", "Green"},
 PlotStyle → {Thick, Dashed, Dotted},
 AxesLabel → {"x (m)", "w(x,t)"},
 PlotLabel → "Comparison at t = 0",
 ImageSize → Large
]

Plot[
 Evaluate[{wA[x, t2], wE[x, t2], wG[x, t2]}],
 {x, 0, l},
 PlotLegends → {"Analytical", "Eigen (10 terms)", "Green"},
 PlotStyle → {Thick, Dashed, Dotted},
 AxesLabel → {"x (m)", "w(x,t)"},
 PlotLabel → "Comparison at t = Pi/(4Ω)",
 ImageSize → Large
]

Plot[
 Evaluate[{wA[x, t3], wE[x, t3], wG[x, t3]}],
 {x, 0, l},
 PlotLegends → {"Analytical", "Eigen (10 terms)", "Green"},
 PlotStyle → {Thick, Dashed, Dotted},
 AxesLabel → {"x (m)", "w(x,t)"},
 PlotLabel → "Comparison at t = Pi/(2Ω)",
 ImageSize → Large
]
```

Out[39]=

Null^2

Syntax: "α[k_] :=" is incomplete; more input is needed.

In[38]:=

```
(*  SOLUTION OF THE ASSIGNMENT 3 BY 22ME10083  *)
(* ==================================================== *)
(* CLEAR WORKSPACE                                       *)
```

```
(* ==================================================== *)

ClearAll[]



(* ==================================================== *)
(* GIVEN DATA                                           *)
(* ==================================================== *)

l  = 2.0;        (* Length (m) *)
d  = 0.005;      (* Diameter (m) *)
ρ  = 7850.0;     (* Density (kg/m^3) *)
T  = 1000.0;     (* Tension (N) *)
Q0 = 60.0;       (* Load amplitude (N) *)
Ω  = 220.0;      (* Excitation frequency (rad/s) *)

A  = Pi d^2/4;      (* Cross-sectional area *)
c  = Sqrt[T/(ρ A)];    (* Wave speed *)
k  = Ω/c;        (* Frequency parameter *)

(* ==================================================== *)
(* (b) ANALYTICAL SOLUTION                              *)
(* ==================================================== *)

Xanalytical[x_] :=
 (Q0/(ρ A Ω^2))*(Cos[k x]/Cos[k l] - 1)

wA[x_, t_] := Xanalytical[x]×Cos[Ω t]

(* ==================================================== *)
(* (c) EIGENFUNCTION EXPANSION (10 TERMS)               *)
(* ==================================================== *)

ω[n_] := ((2 n - 1) Pi c)/(2 l)

W[n_, x_] := Cos[((2 n - 1) Pi x)/(2 l)]

α[n_] :=
 (4 l Q0)/((2 n - 1) Pi ρ A (ω[n]^2 - Ω^2))

XE[x_] :=
 Sum[α[n]×W[n, x], {n, 1, 10}]
```

```
wE[x_, t_] := XE[x] × Cos[Ω t]

(* ==================================================== *)
(* (d) GREEN'S FUNCTION METHOD                           *)
(* ==================================================== *)

G[x_, xb_] :=
 Piecewise[{
  {Cos[k x] × Sin[k (l - xb)] / (k Cos[k l]), x < xb},
  {Cos[k xb] × Sin[k (l - x)] / (k Cos[k l]), x ≥ xb}
 }]

XG[x_] :=
 NIntegrate[G[x, xb]*(Q0 / (ρ A)), {xb, 0, l}]

wG[x_, t_] := XG[x] × Cos[Ω t]

(* ==================================================== *)
(* (e) COMPARISON PLOTS                                  *)
(* ==================================================== *)

t1 = 0;
t2 = Pi / (4 Ω);
t3 = Pi / (2 Ω);

Plot[
 Evaluate[{wA[x, t1], wE[x, t1], wG[x, t1]}],
 {x, 0, l},
 PlotLegends → {"Analytical", "Eigen (10 terms)", "Green"},
 PlotStyle → {Thick, Dashed, Dotted},
 AxesLabel → {"x (m)", "w(x,t)"},
 PlotLabel → "Comparison at t = 0",
 ImageSize → Large
]

Plot[
 Evaluate[{wA[x, t2], wE[x, t2], wG[x, t2]}],
 {x, 0, l},
 PlotLegends → {"Analytical", "Eigen (10 terms)", "Green"},
 PlotStyle → {Thick, Dashed, Dotted},
 AxesLabel → {"x (m)", "w(x,t)"},
```

```
 PlotLabel → "Comparison at t = π/(4Ω)",
 ImageSize → Large
]

Plot[
 Evaluate[{wA[x, t3], wE[x, t3], wG[x, t3]}],
 {x, 0, l},
 PlotLegends → {"Analytical", "Eigen (10 terms)", "Green"},
 PlotStyle → {Thick, Dashed, Dotted},
 AxesLabel → {"x (m)", "w(x,t)"},
 PlotLabel → "Comparison at t = π/(2Ω)",
 ImageSize → Large
]
```

Out[50]=

```
Null^2
```

Syntax: "α[n_] :=" is incomplete; more input is needed.

```
l = 2.0;
rho = 7850.0;
A = 0.00001;
Q0 = 60.0;
Omega = 220.0;
c = 80.0;

omega[n_] := ((2*n - 1)*Pi*c)/(2*l)

alpha[n_] := (2*l*Q0)/((2*n - 1)*Pi*rho*A*(omega[n]^2 - Omega^2))

alpha[1]
```

Out[23]=

```
-0.0218927
```

```
The Green's Function for 0 <= x <= z is:
```

Out[309]//TraditionalForm=

$$\frac{\sec\left(\frac{l\,\Omega}{c}\right)\cos\left(\frac{x\,\Omega}{c}\right)\sin\left(\frac{\Omega\,(l-z)}{c}\right)}{c\,\Omega}$$

```
The Green's Function for z <= x <= l is:
```

Out[311]//TraditionalForm=

$$\frac{\sec\left(\frac{l\,\Omega}{c}\right)\cos\left(\frac{\Omega\,z}{c}\right)\sin\left(\frac{\Omega\,(l-x)}{c}\right)}{c\,\Omega}$$

```
The Green's Function for 0 <= x <= z is:
```

Out[297]//TraditionalForm=

$$\frac{\sec\left(\frac{l\,\Omega}{c}\right)\cos\left(\frac{x\,\Omega}{c}\right)\sin\left(\frac{\Omega\,(l-z)}{c}\right)}{c\,\Omega}$$

```
The Green's Function for z <= x <= l is:
```

Out[299]//TraditionalForm=

$$\frac{\sec\left(\frac{l\,\Omega}{c}\right)\cos\left(\frac{\Omega\,z}{c}\right)\sin\left(\frac{\Omega\,(l-x)}{c}\right)}{c\,\Omega}$$

ASSIGNMENT 3 (FORCED VIBRATION) -- 22ME10083

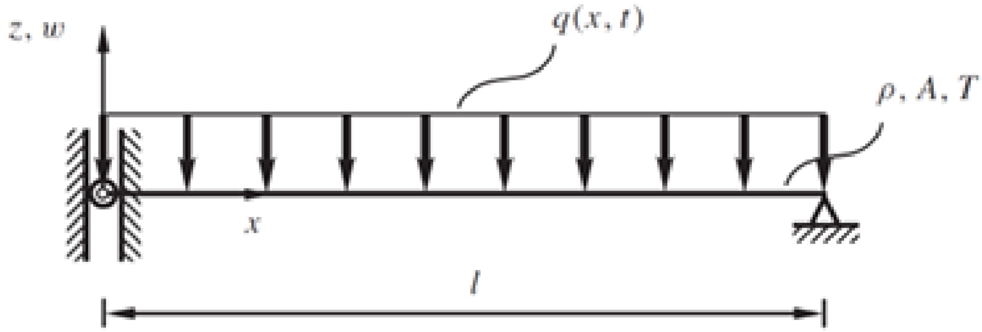


Figure 1: Sliding-fixed string with uniform harmonic load

In[792]:=

```
ClearAll["Global`*"]
(* 1. Define the Piecewise Green's Function forms *)
GL[x_, z_, Ω_] := AL * Sin[Ω * x / c] + BL * Cos[Ω * x / c];
GR[x_, z_, Ω_] := AR * Sin[Ω * x / c] + BR * Cos[Ω * x / c];

(* 2. Define the Boundary and Continuity/Jump Conditions *)
eq1 = (D[GL[x, z, Ω], x] /. x → 0) == 0;
eq2 = GR[l, z, Ω] == 0;
eq3 = GL[z, z, Ω] == GR[z, z, Ω];
eq4 = (D[GR[x, z, Ω], x] - D[GL[x, z, Ω], x] /. x → z) == -1/c^2;

(* 3. Solve for the coefficients *)
sol = Simplify[Solve[{eq1, eq2, eq3, eq4}, {AL, BL, AR, BR}]];

(* 4. Display the results in Symbolic/Traditional Form *)
Print[];
GL[x, z, Ω] /. sol[[1]] // Simplify // TraditionalForm

Print["The Green's Function for z <= x <= l is:"];
GR[x, z, Ω] /. sol[[1]] // Simplify // TraditionalForm

(* Extract and Display the simplified expressions *)
finalGL = Simplify[GL[x, z, Ω] /. sol[[1]]];
finalGR = Simplify[GR[x, z, Ω] /. sol[[1]]];
Print["Simplified GL: ", finalGL];
Print["Simplified GR: ", finalGR];
(* Solve for Displacement using Green's Function Integration *)
xpg = (-Q0 / (ρ * a * l)) * (Integrate[finalGR, {z, 0, x}] +
        Integrate[finalGL, {z, x, l}]);
wpG[x_, t_] = xpg * Cos[Ω * t];
wpG[x, t] // Simplify
```

The Green's Function for 0 <= x <= z is:

Out[801]//TraditionalForm=

$$\frac{\sec\left(\frac{l\,\Omega}{c}\right)\cos\left(\frac{x\,\Omega}{c}\right)\sin\left(\frac{\Omega\,(l-z)}{c}\right)}{c\,\Omega}$$

The Green's Function for z <= x <= l is:

Out[803]//TraditionalForm=

$$\frac{\sec\left(\frac{l\,\Omega}{c}\right)\cos\left(\frac{\Omega\,z}{c}\right)\sin\left(\frac{\Omega\,(l-x)}{c}\right)}{c\,\Omega}$$

Simplified GL: $\dfrac{\text{Cos}\left[\frac{x\,\Omega}{c}\right] \times \text{Sec}\left[\frac{l\,\Omega}{c}\right] \times \text{Sin}\left[\frac{(l-z)\,\Omega}{c}\right]}{c\,\Omega}$

Simplified GR: $\dfrac{\text{Cos}\left[\frac{z\,\Omega}{c}\right] \times \text{Sec}\left[\frac{l\,\Omega}{c}\right] \times \text{Sin}\left[\frac{(l-x)\,\Omega}{c}\right]}{c\,\Omega}$

Out[810]=

$$-\frac{2\,Q0\,\text{Cos}[t\,\Omega] \times \text{Sec}\left[\frac{l\,\Omega}{c}\right] \times \text{Sin}\left[\frac{(l-x)\,\Omega}{2\,c}\right] \times \text{Sin}\left[\frac{(l+x)\,\Omega}{2\,c}\right]}{a\,l\,\rho\,\Omega^2}$$

MAIN LOGIC TO SOLVE THE PROBLEM

In[820]:=

```
(* Initial Clear and Parameters *)
Clear[];
Q0 = 60; ρ = 7860; a = (Pi*0.25*0.005^2); l = 2; T = 1000;

(* Characteristic constants *)
c = Sqrt[T/(ρ * a)];
Ω = 220; (* Forcing Frequency rad/s *)
num = 10; (* Number of terms for expansion *)


(* --- 0.  Calculating the Steady state solution wA(x,t) --- *)
wA[x_, t_] = -(Q0/(ρ*a*l*Ω ^2))*((Cos[Ω*x/c]/Cos[Ω*l/c]) - 1)*Cos[Ω*t];
wA[x, t]


(* --- 1. Eigenfunction Expansion Method --- *)
Print[Style[, 12, Black, Bold]];

W[k_] := Cos[(2*k - 1) * Pi * x /(2* l)];
ω[k_] := (2*k - 1) * Pi * c / (2*l);


(* Modal coefficient α *)
α[k_] := Integrate[(-Q0/(ρ * a*l)) * W[k], {x, 0, l}] /
    ((-Ω^2 + ω[k]^2) * Integrate[W[k]^2, {x, 0, l}]);
  α[k]

(* Summing the expansion *)
xpmd = Sum[α[k] * W[k], {k, 1, num}];

(* General expression used for comparison *)
```

```
wpE[x_, t_] := xpmd * Cos[Ω * t]
wpE[x, t]


(*(* --- 2. Exact Analytical Solution --- *)
Print[Style[, 18, Red, Bold]];

(* Solving the Boundary Value Problem (BVP) using DSolve *)
xpsol = X[x] /. Flatten[DSolve[
    {X''[x] + (Ω^2 / c^2) * X[x] == -f0/T, X[0] == 0, X[l] == 0},
    X[x], x]];

(* Manual Analytical Expression for comparison *)
xpe = (-f0 / (ρ * a * Ω^2)) +
      (f0 / (ρ * a * Ω^2)) * Cos[Ω * x / c] +
      ((f0 * (1 - Cos[Ω * l / c])) / (ρ * a * Ω^2 * Sin[Ω * l / c])) * Sin[Ω * x / c]; *)


(* --- 3. Green's Function Method --- *)
Print[Style[, 12, Black, Bold]];

(* Define Piecewise Green's Function forms *)
(* 1. Define the Piecewise Green's Function forms *)
GL[x_, z_, Ω_] := AL * Sin[Ω * x / c] + BL * Cos[Ω * x / c];
GR[x_, z_, Ω_] := AR * Sin[Ω * x / c] + BR * Cos[Ω * x / c];
(* 2. Define the Boundary and Continuity/Jump Conditions *)
(* eq1: Zero slope at x=0 (Free End) *)
eq1 = (D[GL[x, z, Ω], x] /. x → 0) == 0;
(* eq2: Fixed at x=l *) eq2 = GR[l, z, Ω] == 0;
(* eq3: Continuity at the point of application x=z *)
eq3 = GL[z, z, Ω] == GR[z, z, Ω];
(* eq4: Slope Jump Condition at x=z *)
eq4 = (D[GR[x, z, Ω], x] - D[GL[x, z, Ω], x] /. x → z) == -1/c^2;
(* 3. Solve for the coefficients *)
sol = Simplify[Solve[{eq1, eq2, eq3, eq4}, {AL, BL, AR, BR}]];

(* 4. Display the results in Symbolic/Traditional Form *)
Print[];
GL[x, z, Ω] /. sol[[1]] // Simplify // TraditionalForm
Print["The Green's Function for z <= x <= l is:"];
GR[x, z, Ω] /. sol[[1]] // Simplify // TraditionalForm
```

```
(* Extract and Display the simplified expressions *)
finalGL = Simplify[GL[x, z, Ω] /. sol[[1]]];
finalGR = Simplify[GR[x, z, Ω] /. sol[[1]]];
Print["Simplified GL: ", finalGL];
Print["Simplified GR: ", finalGR];
(* Solve for Displacement using Green's Function Integration *)
xpg = (-Q0 / (ρ * a*l)) * (Integrate[finalGR, {z, 0, x}] +
         Integrate[finalGL, {z, x, l}]);

wpG[x_, t_] = xpg*Cos[Ω*t];
wpG[x, t]

(* --- 4. Comparison Plot --- *)
Print[Style[, 18, Red, Bold]];

Plot[wA[x, 0.1], {x, 0, l }]
Plot[wpE[x, 0.1], {x, 0, l }]
Plot[wpG[x, 0.1], {x, 0, l }]
```

Out[826]=

$$-0.00401628\, \text{Cos}[220\, t]\, (-1 + 1.46122\, \text{Cos}[2.73306\, x])$$

Eigenfunction expansion method

Out[831]=

$$-\frac{247.502\, \text{Cos}[k\,\pi]}{(1. - 2.\, k)\left(-48\,400 + 3996.94\,(-1 + 2\,k)^2\right)\left(1 + \frac{\text{Sin}[2\,k\,\pi]}{\pi - 2\,k\,\pi}\right)}$$

Out[834]=

$$\text{Cos}[220\, t]\Big(0.00557399\, \text{Cos}\Big[\frac{\pi x}{4}\Big] - 0.00663853\, \text{Cos}\Big[\frac{3\pi x}{4}\Big] - 0.000960736\, \text{Cos}\Big[\frac{5\pi x}{4}\Big] + 0.000239793\, \text{Cos}\Big[\frac{7\pi x}{4}\Big] - 0.0000998731\, \text{Cos}\Big[\frac{9\pi x}{4}\Big] + 0.0000516973\, \text{Cos}\Big[\frac{11\pi x}{4}\Big] - 0.0000303606\, \text{Cos}\Big[\frac{13\pi x}{4}\Big] + 0.0000193912\, \text{Cos}\Big[\frac{15\pi x}{4}\Big] - 0.0000131551\, \text{Cos}\Big[\frac{17\pi x}{4}\Big] + 9.34133\times 10^{-6}\, \text{Cos}\Big[\frac{19\pi x}{4}\Big]\Big)$$

Green's function method

The Green's Function for 0 <= x <= z is:

Out[844]//TraditionalForm=

$$\cos(2.73306\, x)\,(-0.0000564681\, \sin(2.73306\, z) - 0.0000601632\, \cos(2.73306\, z))$$

The Green's Function for z <= x <= l is:

Out[846]//TraditionalForm=

$$\cos(2.73306\,z)\,(-0.0000564681\,\sin(2.73306\,x) - 0.0000601632\,\cos(2.73306\,x))$$

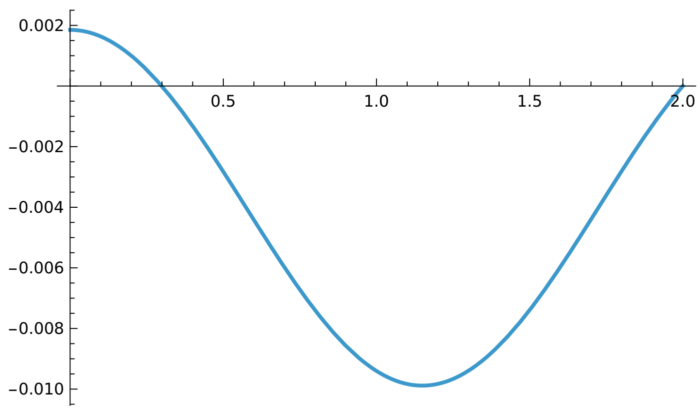
```
Simplified GL: Cos[2.73306 x] (-0.0000601632 Cos[2.73306 z] - 0.0000564681 Sin[2.73306 z])
Simplified GR: Cos[2.73306 z] (-0.0000601632 Cos[2.73306 x] - 0.0000564681 Sin[2.73306 x])
```

Out[853]=

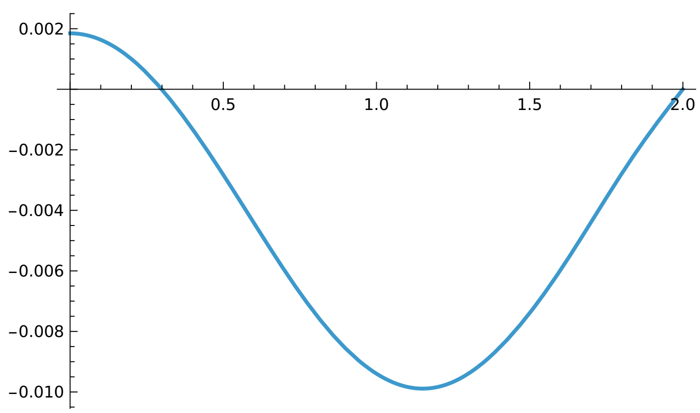
```
-194.388 Cos[220 t]
 (Cos[2.73306 x] (0.0000301904 - 0.0000206612 Cos[2.73306 x] + 0.0000220131 Sin[2.73306 x]) -
   0.0000206612 Sin[2.73306 x]^2 - 0.0000110066 Sin[5.46612 x])
```

Comparison of all three methods

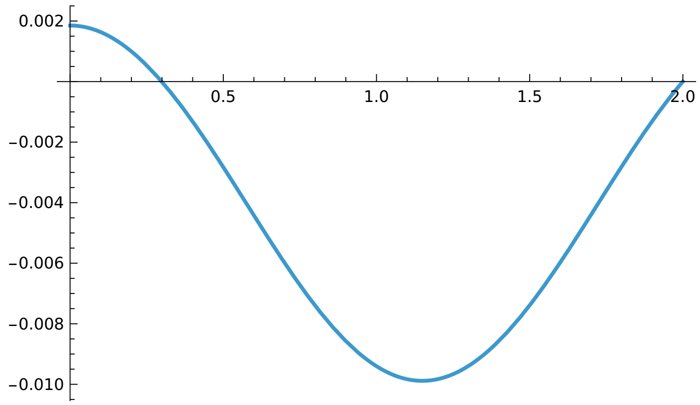
Out[855]=



Out[856]=



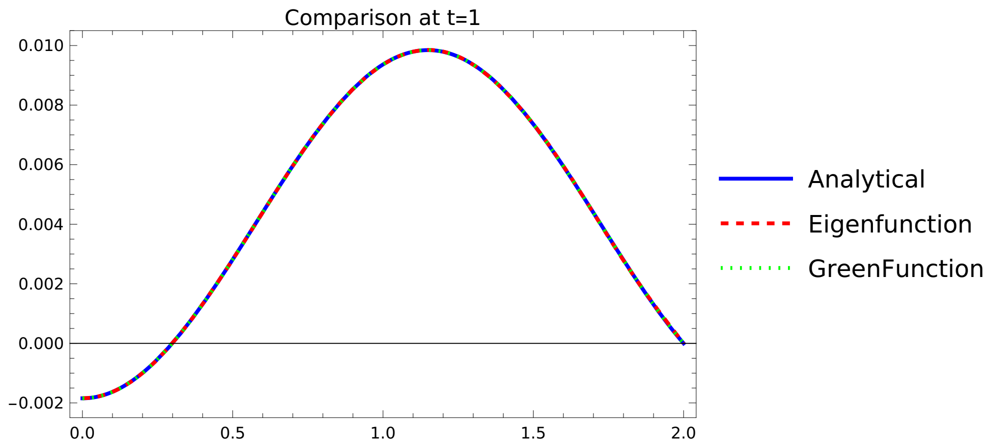
Out[857]=



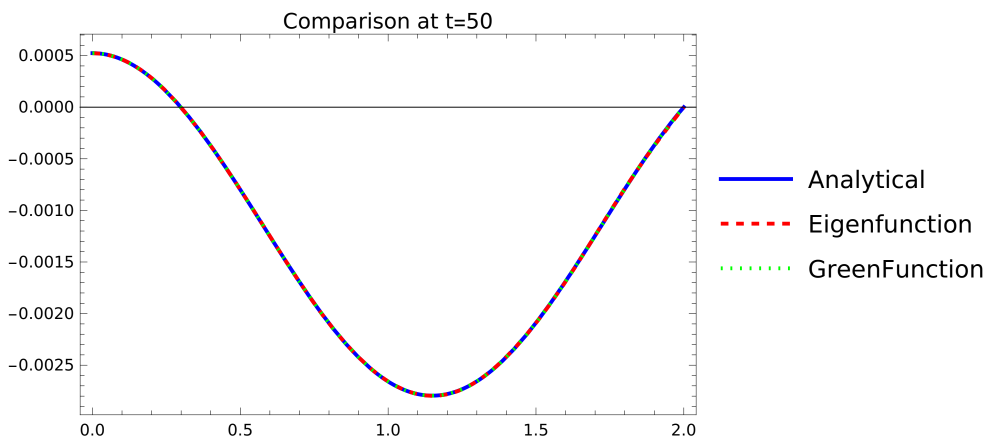
In[858]:=

```
(* Comparison Plot at t=1 *)
Plot[{wA[x, 1], wpE[x, 1], wpG[x, 1]}, {x, 0, l},
PlotStyle → {Blue, {Red, Dashed}, {Green, Dotted}},
PlotLegends → {, , },
Frame → True, PlotLabel → ]


Plot[{wA[x, 50], wpE[x, 50], wpG[x, 50]}, {x, 0, l},
PlotStyle → {Blue, {Red, Dashed}, {Green, Dotted}},
PlotLegends → {"Analytical", "Eigenfunction", "GreenFunction"},
Frame → True, PlotLabel → "Comparison at t=50"]


Plot[{wA[x, 700], wpE[x, 700], wpG[x, 700]}, {x, 0, l},
PlotStyle → {Blue, {Red, Dashed}, {Green, Dotted}},
PlotLegends → {"Analytical", "Eigenfunction", "GreenFunction"},
Frame → True, PlotLabel → "Comparison at t=700"]
```

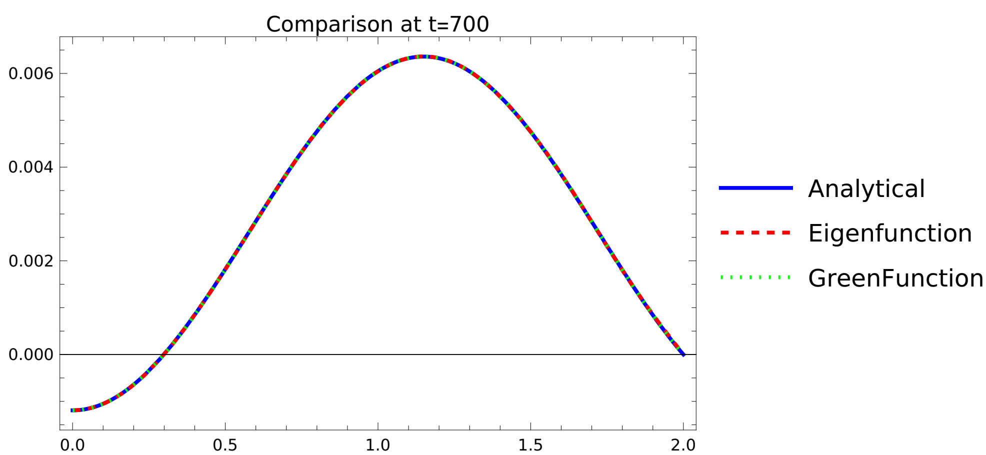
Out[858]=



Out[859]=



Out[860]=



Plot: Limiting value l in {x, 0, l} is not a machine–sized real number.

Out[814]=

```
Plot[{wA[x, 1], wpE[x, 1], wpG[x, 1]}, {x, 0, l}, PlotStyle → {Blue, {Red, Dashed}, {Green, Dotted}},
 PlotLegends → {Analytical, Eigenfunction, GreenFunction},
 Frame → True, PlotLabel → Comparison at t=1]
```

Plot: Limiting value l in {x, 0, l} is not a machine–sized real number.

Out[815]=

```
Plot[{wA[x, 50], wpE[x, 50], wpG[x, 50]},
 {x, 0, l}, PlotStyle → {Blue, {Red, Dashed}, {Green, Dotted}},
 PlotLegends → {Analytical, Eigenfunction, GreenFunction},
 Frame → True, PlotLabel → Comparison at t=50]
```

Plot: Limiting value l in {x, 0, l} is not a machine-sized real number.

Out[816]=

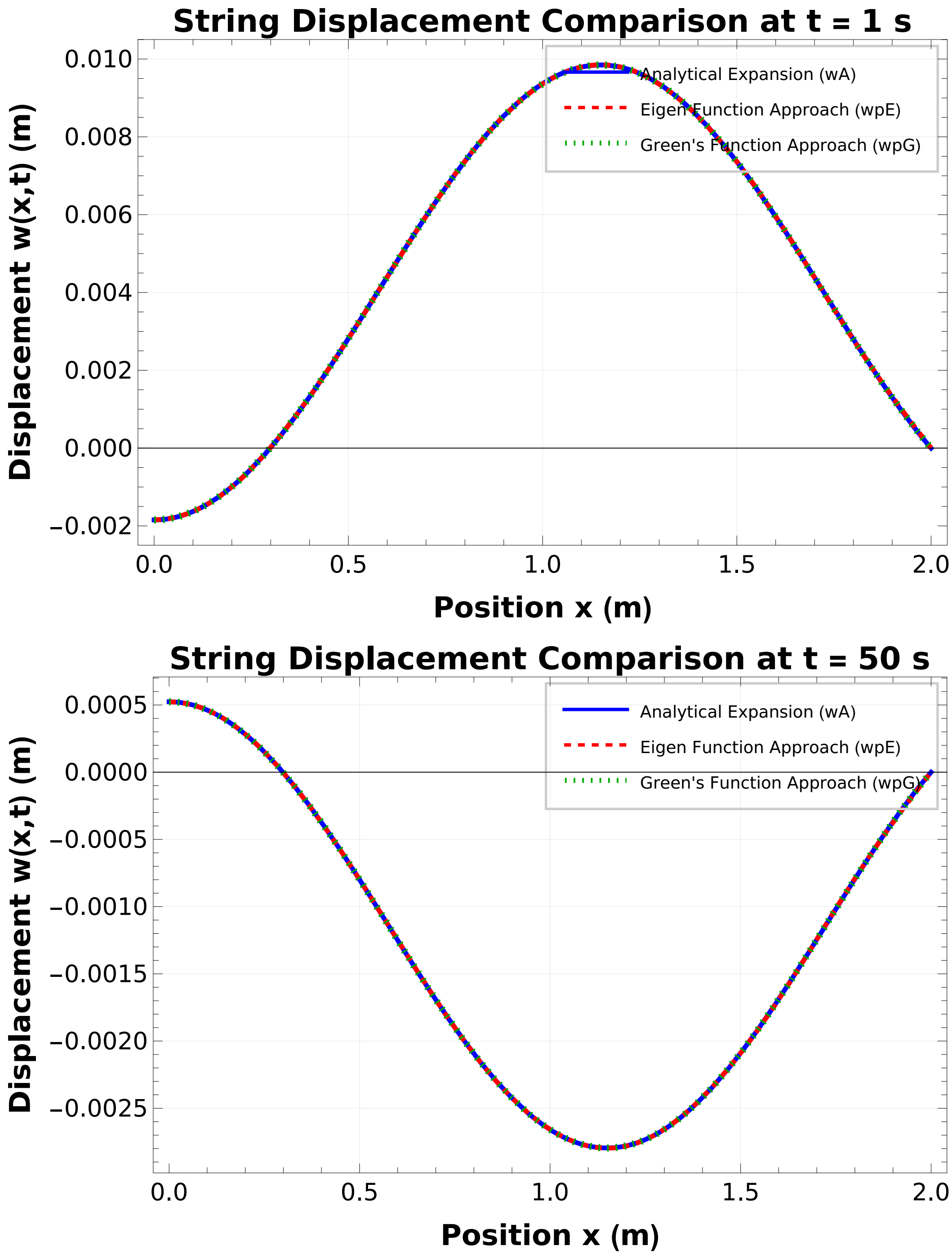
```
Plot[{wA[x, 700], wpE[x, 700], wpG[x, 700]},
 {x, 0, l}, PlotStyle → {Blue, {Red, Dashed}, {Green, Dotted}},
 PlotLegends → {Analytical, Eigenfunction, GreenFunction},
 Frame → True, PlotLabel → Comparison at t=700]
```

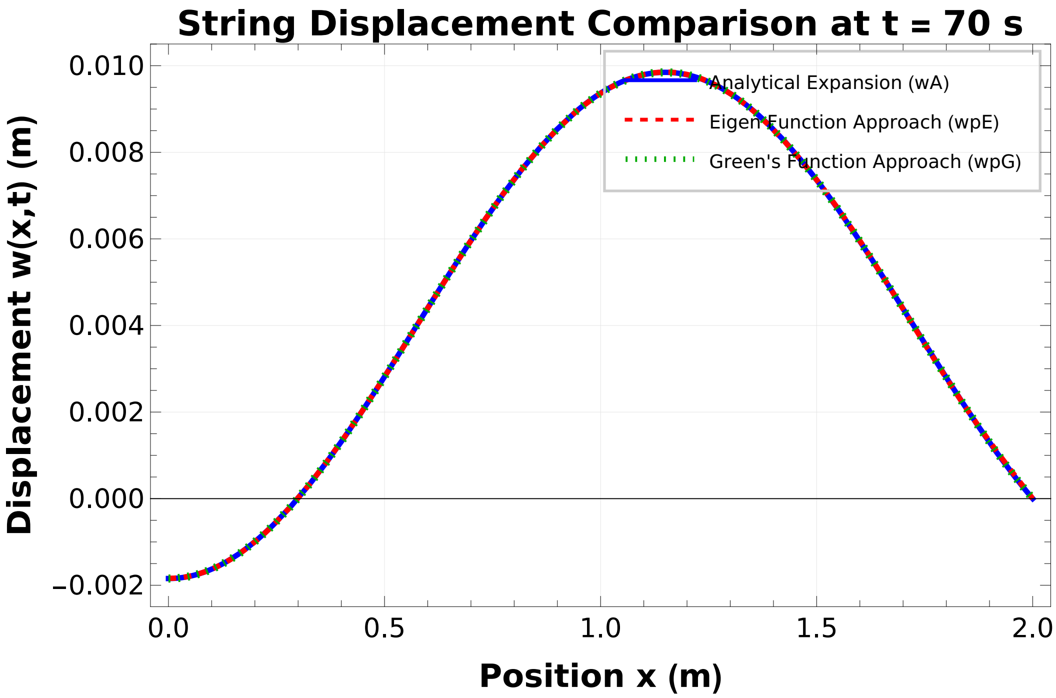
VIZUALIZATION

In[873]:=

```
(* Define a list of times for the plots *)
plotTimes = {1, 50,  70};



(* Define specific styles for publication clarity *)
lineStyles = {
 {Blue, Thickness[0.006]},        (* wA: Solid Blue *)
 {Red, Dashed, Thickness[0.006]},     (* wpE: Dashed Red *)
 {Darker[Green], Dotted, Thickness[0.009]} (* wpG: Dotted Green *)
};



(* Generate the plots *)
Do[
 Print[
  Plot[{wA[x, tVal], wpE[x, tVal], wpG[x, tVal]}, {x, 0, l},

   (* Line Styles and Colors *)
   PlotStyle → lineStyles,

   (* Frame and Labeling *)
   Frame → True,
   FrameStyle → Directive[Black, 14],
   GridLines → Automatic,
   GridLinesStyle → Directive[GrayLevel[0.9], Thin],
```

```
(* Labels *)
FrameLabel → {
 Style[, 16, Bold],
 Style[, 16, Bold]
},
PlotLabel → Style[ <> ToString[tVal] <> , 18, Bold],

(* Legends: High Quality Placement *)
PlotLegends → Placed[
 LineLegend[
  {, , },
  LegendFunction → (Framed[#, FrameStyle → GrayLevel[0.8]] &),
  BaseStyle → {FontSize → 12, Bold}
 ],
 {Right, Top}
],

(* Image Sizing for Publications *)
ImageSize → 550,
PlotRange → All,
AspectRatio → 1/1.6 (* Golden ratio approximation *)
]
],
{tVal, plotTimes}
]
```

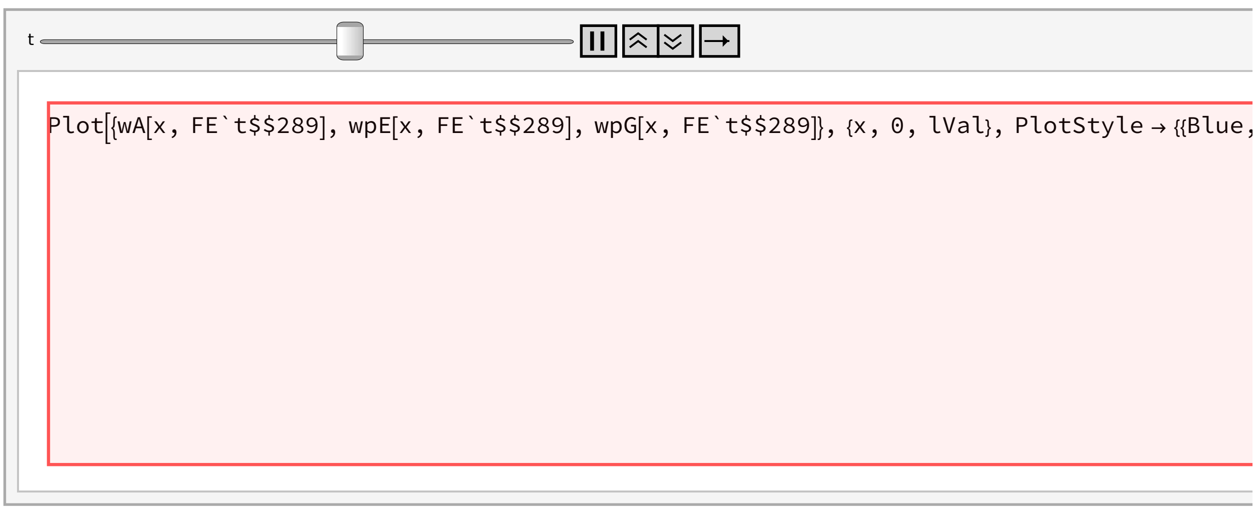

String Displacement Comparison at t = 1 s
Displacement w(x,t) (m)
Position x (m)
Analytical Expansion (wA)
Eigen Function Approach (wpE)
Green's Function Approach (wpG)
0.010
0.008
0.006
0.004
0.002
0.000
−0.002
0.0
0.5
1.0
1.5
2.0
String Displacement Comparison at t = 50 s
Displacement w(x,t) (m)
Position x (m)
Analytical Expansion (wA)
Eigen Function Approach (wpE)
Green's Function Approach (wpG)
0.0005
0.0000
−0.0005
−0.0010
−0.0015
−0.0020
−0.0025
0.0
0.5
1.0
1.5
2.0



String Displacement Comparison at t = 70 s
Analytical Expansion (wA)
Eigen Function Approach (wpE)
Green's Function Approach (wpG)
Displacement w(x,t) (m)
Position x (m)
0.010
0.008
0.006
0.004
0.002
0.000
−0.002
0.0
0.5
1.0
1.5
2.0

In[699]:=
```
(* 1. Setup specific styles for publication-quality animation *)
(* 3. Animation using ONLY shared variables *)
Animate[Plot[{wA[x, t], wpE[x, t], wpG[x, t]},
  {x, 0, lVal}, (* Styling based on your specific requirements *)
  PlotStyle → {{Blue, Thickness[0.006]}, (* wA *){Red, Dashed, Thickness[0.006]},
    (* wpE *){Darker[Green], Dotted, Thickness[0.009]}(* wpG *)},
  (* Fixed Axis for Stability *)PlotRange → {All, {-0.015, 0.015}},
  Frame → True, GridLines → Automatic, (* Labels & Legends *)
  FrameLabel → {Style[, 14], Style[, 14]}, PlotLabel → Style[, 16, Bold],
  PlotLegends → Placed[LineLegend[{, , }, LegendFunction →
      (Framed[#, FrameStyle → GrayLevel[0.8]] &)], Below], ImageSize → 550],
 {t, 0, 10, 0.05}, (* Time slider from 0 to 10 seconds *)AnimationRate → 0.5]
```

Out[699]=



In[51]:=
```
(* Initial Clear and Parameter Definitions *)
Clear[];



num = 21;  (* Number of terms in the expansion *)
l = 1;    (* Length of the string *)
a = 1;    (* Cross-sectional area *)
ρ = 1;    (* Density *)
T = 1;    (* Tension *)
f = 1;    (* Magnitude of the moving force *)

(* Derived Physical Constants *)
c = Sqrt[T / (ρ * a)]; (* Wave speed *)
v = 0.1 * c;    (* Velocity of the moving force *)
```

```
(* Calculation of Natural Frequencies *)
ω = Table[i * Pi * c / l, {i, 1, num}];

(* --- Complete Analytical Solution for Field Variable w(x, t) --- *)
(* This represents the string displacement with zero initial conditions *)

w[x_, t_] := (-(2 * f * l) / (ρ * a * c * Pi^2 * (c^2 - v^2))) * Sum[
  (1/j^2) * (
   Sin[(j * Pi * v * t) / l] -
   (v/c) * Sin[(j * Pi * c * t) / l]
  ) * Sin[(j * Pi * x) / l],
  {j, 1, num}
 ];

(* --- Dynamic Visualization --- *)
(* Animates the string displacement as the force moves across it *)

DynamicModule[{n = 0},
 Column[{
  (* Animator control for time (t) from 0 to l/v *)
  Animator[Dynamic[n], {0, l/v}, 0.4],

  Dynamic[
   Plot[
    w[x, n], {x, 0, l},
    (* Epilog adds a blue point representing the moving load location *)
    Epilog → {
     Blue, PointSize[0.03],
     Point[{n * v, w[n * v, n]}]
    },
    ImageSize → 400,
    PlotRange → {{0, l}, {-0.4, 0.4}},
    Frame → True,
    PlotStyle → {Thick, Red},
    GridLines → Automatic,
    AspectRatio → 4/6,
    FrameLabel → {
     {Style[, 16], None},
     {Style[, 16], Style[, 16]}
    }
   ]
```

```
  ],
  (* Display current time *)
  Dynamic[Row[{, n, }]]
 }]
]
```

Out[62]=

```
Plot[w[x, FE`n$$290], {x, 0, l},
 Epilog → {Blue, PointSize[0.03], Point[{FE`n$$290 v, w[FE`n$$290 v, FE`n$$290]}]},
 ImageSize → 400, PlotRange → {{0, l}, {-0.4, 0.4}}, Frame → True, PlotStyle → {Thick, Red},
 GridLines → Automatic, AspectRatio → 4/6, FrameLabel → {{w(x, t), None},
   {length, x, String with uniform moving load F δ(x-vt)}}]
```

Time: 1.2792 secs